

MATH 345 Skeleton Notes

Unit 1: Vectors, matrices, and linear maps

Section 1.1: Vectors

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Vectors as lists of numbers

A real quantity is also called a .

Definition: Vector An is an ordered list of n scalars, often written in the form $\mathbf{v} = (v_1, v_2, \dots, v_n)$. The collection of all n -vectors is denoted $\mathbb{R}^n$.

We will also encounter vectors written as a row or column of numbers in square brackets, i.e.,

$$[v_1 \ \cdots \ v_n] \quad \text{and} \quad \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}.$$

It is customary to use bold letters for variables, such as $\mathbf{v}$ or $\mathbf{w}$, to represent vectors. We let $\mathbf{0}$ denote the vector $(0, \dots, 0)$ consisting only of zeros. A point in n -dimensional space can also be identified with an ordered list of n numbers using Cartesian coordinates. When referring specifically to a point P , we write $P(x_1, x_2, \dots, x_n)$ to indicate its coordinates.

Vectors as displacements in space

Vectors can be used to represent displacements between points in space. The vector $\mathbf{v} = (v_1, v_2, v_3)$ represents the displacement from $P(x_1, y_1, z_1)$ to

$Q(x_2, y_2, z_2)$, where $x_2 = x_1 + v_1$, $y_2 = y_1 + v_2$, and $z_2 = z_1 + v_3$. This vector is also written as $\overrightarrow{PQ}$. Because vectors often indicate displacements, they are drawn pictorially as arrows starting at a point, and ending where that point is displaced by the vector.

Activity: Vectors as displacements in space

What is the vector representing the displacement from the point $P(1, 2)$ to the point $Q(5, 7)$.

Definition: Length of a vector The of a vector $\mathbf{v} = (v_1, \dots, v_n)$ is the quantity

$$\|\mathbf{v}\| = \sqrt{v_1^2 + \dots + v_n^2}.$$

Activity

Verify, using planar geometry, that the length of a 2-vector is the length of the line-segment of the arrow representing the vector.

In a calculus course, you may have mostly seen a vector as a displace-

ment, but in this course it can also be a row of data or a list of model parameters, among other important applications. The same operations below will later compare documents, tokens, and feature vectors.

Adding and scaling vectors

Two displacements can be combined: just apply one displacement after the other is applied. This naturally leads to the notion of *adding* two vectors.

Definition: Adding vectors The sum of two n -vectors $\mathbf{v} = (v_1, \dots, v_n)$ and $\mathbf{w} = (w_1, \dots, w_n)$, denoted $\mathbf{v} + \mathbf{w}$, is the vector given by the tuple $(v_1 + w_1, \dots, v_n + w_n)$.

Remark The sum of two vectors can be illustrated pictorially as in textbook figure fig-1-1-adding-vectors by drawing the parallelogram with the vectors placed along two sides of the parallelogram.

One can also *scale* a vector by a given scalar quantity.

Definition: Scaling vectors If $\mathbf{v} = (v_1, \dots, v_n)$ is an n -vector and $t \in \mathbb{R}$ is a scalar, we let $t\mathbf{v}$ denote the vector $(tv_1, \dots, tv_n)$, i.e., multiplying each entry of the vector $\mathbf{v}$ by t .

Definition: Linear combinations and convex combinations of vectors If $\mathbf{v}_1, \dots, \mathbf{v}_k$ are vectors in $\mathbb{R}^n$ and $c_1, \dots, c_k$ are scalars, then

$$c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k$$

is a

of $\mathbf{v}_1, \dots, \mathbf{v}_k$. The scalars $c_1, \dots, c_k$ are

the

of the linear combination. If $c_i \geq 0$ for all i

and $c_1 + \dots + c_k = 1$, then the linear combination is a

.

A convex combination can be interpreted as a weighted average.

Activity: A weighted average

Let $\mathbf{v}_1 = (10, 0)$, $\mathbf{v}_2 = (0, 10)$, $\mathbf{v}_3 = (10, 10)$, and $\alpha = (1/4, 1/4, 1/2)$. Compute $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3$.

Dot products of vectors

Definition: Dot product The of two n -vectors $\mathbf{v} = (v_1, \dots, v_n)$ and $\mathbf{w} = (w_1, \dots, w_n)$ is the scalar quantity

$$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + \dots + v_n w_n.$$

Geometrically, the dot product of two vectors is a quantity which is related to the *angle* between the two vectors $\mathbf{v}$ and $\mathbf{w}$. If we draw the two vectors as arrows with the same starting point, then they form an angle on the plane containing both vectors, and the angle θ is given below.

Theorem: Angle between vectors For two nonzero vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$, the θ between the vectors is the quantity satisfying the equation

$$\cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|}.$$

The dot product can be thought of as a measure of the *similarity* between two vectors. Suppose θ is the angle between two vectors $\mathbf{v}$ and $\mathbf{w}$:

- If θ is close to zero, then the two vectors are close to pointing in the same direction. Since $\cos(0^\circ) = 1$, this occurs precisely when

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} \approx 1.$$

- If θ is close to 180° , then the two vectors are close to pointing in opposite directions. Since $\cos(180^\circ) = -1$, this occurs precisely when

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} \approx -1.$$

- If θ is close to 90° , the two vectors are close to pointing at right angles. Since $\cos(90^\circ) = 0$, this occurs precisely when

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} \approx 0.$$

When $\mathbf{v} \cdot \mathbf{w} = 0$, so that $\theta = 90^\circ$, we say the two vectors are perpen-

dicular, or .

Thus the quantity

$$\frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|}$$

ranges between -1 and 1 , and measures the degree to which the two vectors $\mathbf{v}$ and $\mathbf{w}$ point in the same, or opposite, directions. In some contexts, especially in certain areas of data science, this quantity is called the

between the vectors $\mathbf{v}$ and $\mathbf{w}$.

Warning Cosine similarity is undefined if either vector is the zero vector.

Activity

Calculate the angle between the two vectors $\mathbf{v} = (1, 2)$ and $\mathbf{w} = (\sqrt{3} + 2, 2\sqrt{3} - 1)$.

Activity

Calculate the angle between the two vectors

$$\mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ 0 \\ 2 \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ -1 \end{bmatrix}.$$

Note Note that $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$ for any $\mathbf{v} \in \mathbb{R}^n$.

We will study the dot product in far more detail in Orthogonality and least squares and Abstract vector spaces and polynomial approximation.

Definition: Distance between vectors The



between two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is

$$\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

It measures how far apart the endpoints are, while cosine similarity compares direction after normalization.

Applications of vectors

Vectors can represent quantities other than displacements. The entries of a vector can record measurements, counts, samples, or features. The meaning of a vector depends on what its coordinates represent.

- **Object:** Color
Meaning of the entries: red, green, and blue intensities
- **Object:** Time series
Meaning of the entries: measurements at successive times
- **Object:** Portfolio
Meaning of the entries: amounts held in each asset
- **Object:** Image
Meaning of the entries: pixel intensities, listed in a fixed order
- **Object:** Document
Meaning of the entries: counts of selected words
- **Object:** Customer
Meaning of the entries: purchases of selected products
- **Object:** Object with features
Meaning of the entries: measured attributes such as size, price, weight, or rating

The same vector operations can have different interpretations. A sum can add purchases, add word counts, or add two time series. A scalar multiple can rescale an image, double a portfolio, or change units. A dot product can produce a score. A distance can compare two feature vectors. Cosine similarity compares direction after normalization.

Activity: Word-count vectors

Use the dictionary `linear, matrix, data`. A document vector records the number of times these words appear, in this order. The query

$$\mathbf{q} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

represents the phrase “linear data”. Consider three document vectors

$$\mathbf{D}_1 = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \quad \mathbf{D}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{D}_3 = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}.$$

Compute the cosine similarities and Euclidean distances between $\mathbf{q}$ and each $\mathbf{D}_i$. Rank the documents by cosine similarity.

From the previous activity, we see that vectors can have high cosine similarity while being relatively far in terms of Euclidean distance.

Activity: The same ranking in code

The following code repeats the cosine-similarity ranking from Word-count vectors.

```
import numpy as np

D1 = np.array([2, 0, 2], dtype=float)
D2 = np.array([1, 1, 0], dtype=float)
D3 = np.array([0, 2, 0], dtype=float)
q = np.array([1, 0, 1], dtype=float)

score1 = (D1 @ q) / (np.linalg.norm(D1) * np.linalg.norm(q))
score2 = (D2 @ q) / (np.linalg.norm(D2) * np.linalg.norm(q))
score3 = (D3 @ q) / (np.linalg.norm(D3) * np.linalg.norm(q))

score1, score2, score3
```

Output:

```
(1.0, 0.5, 0.0)
```

1. Which score corresponds to $\mathbf{D}_2$?
2. Where does the code compute $\mathbf{D}_1 \cdot \mathbf{q}$?
3. Which document is most similar to the query by cosine similarity?
4. Why does the output agree with the word-count vector activity?

Example: Neural networks and transformers A neural network is a function built from layers. A layer takes numbers as input, combines them using weights, applies a rule, and sends output numbers to the next layer. In many neural networks, the input and output of a layer are vectors. The weights are adjusted from data during training.

A transformer is a neural-network architecture for sequences. In a language model, text is first broken into tokens. A token can be a word, part of a word, punctuation mark, or other text fragment. Each token is represented by a vector.

Activity: A tiny attention-style token-weighting calculation

Consider the token sequence

small, red, bird.

Suppose a transformer layer is updating the token “bird”. Its

vector represents what this token is looking for in the sequence. Each vector represents how another token can be matched by a query. Each vector is the vector that may be averaged into the updated representation.

For this activity, the score for token i is

$$\text{score}_i = \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_i.$$

Use

$$\mathbf{q}_{\text{bird}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{k}_{\text{small}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{k}_{\text{red}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{k}_{\text{bird}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

1. Compute the three scores

$$\mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{small}}, \quad \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{red}}, \quad \mathbf{q}_{\text{bird}} \cdot \mathbf{k}_{\text{bird}}.$$

2. Normalize the positive scores by dividing each score by the sum of all three scores. Call the resulting weights

$$\alpha = \begin{bmatrix} \alpha_{\text{small}} \\ \alpha_{\text{red}} \\ \alpha_{\text{bird}} \end{bmatrix}.$$

3. Use the value vectors

$$\mathbf{v}_{\text{small}} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \quad \mathbf{v}_{\text{red}} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \quad \mathbf{v}_{\text{bird}} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$



Warning This toy calculation uses normalization by the sum of positive scores to keep the arithmetic simple. Real transformer attention usually uses a different weighting rule.